

# BioSpecInfo

## Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

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# Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

| Field    | Value                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Software | BioSpecInfo                                                                                                                             |
| Author   | Samuele Pio Provenzano                                                                                                                  |
| Purpose  | Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations. |

## 1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

## 2. Software verification (QA)

| Technique                               | Description                                                                                                                                                                                                                                                                              | Status                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Syntax checking                         | <code>node --check</code> on the critical JavaScript blocks on every change.                                                                                                                                                                                                             | ✓ In use                                                                                      |
| End-to-end (E2E) testing                | Automation with <b>Playwright</b> on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: <b>0 non-network errors</b> ). | ✓ In use                                                                                      |
| HTTP-server verification                | RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.                                                                                                                                                                               | ✓ In use                                                                                      |
| Manual cross-device testing             | Testing on real devices (see §4).                                                                                                                                                                                                                                                        | ✓ In use                                                                                      |
| Automated unit-test suite (Vitest/Jest) | —                                                                                                                                                                                                                                                                                        |  Roadmap |
| Coverage ≥ 90%                          | —                                                                                                                                                                                                                                                                                        |  Roadmap |

Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

### 3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

#### 3.1 <sup>1</sup>H-NMR benchmark vs SDBS

| Metric                       | Reported value                                              |
|------------------------------|-------------------------------------------------------------|
| Reference database           | <b>SDBS</b> (Spectral Database for Organic Compounds, AIST) |
| Mean Absolute Error (MAE)    | <b>0.31 ppm</b>                                             |
| Signals within $\pm 0.5$ ppm | <b>89%</b>                                                  |

#### 3.2 IR benchmark (C=O band) vs NIST

| Metric                                 | Reported value                |
|----------------------------------------|-------------------------------|
| Reference database                     | <b>NIST Chemistry WebBook</b> |
| Mean Absolute Error (MAE)              | <b>11 cm<sup>-1</sup></b>     |
| Bands within $\pm 30$ cm <sup>-1</sup> | <b>90%</b>                    |

#### 3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol*  $\rightarrow$  *C<sub>27</sub>H<sub>46</sub>O*, 4 rings, 8 stereocentres.

**Benchmark data source:** measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

### 4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

| Device                 | Result    |
|------------------------|-----------|
| iPhone / iPad          | 30-60 fps |
| Windows / macOS laptop | 60 fps    |
| Raspberry Pi 4         | 30-60 fps |

Size of the main `index.html` file:  $\approx$  **4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

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## 5. Declared model limitations

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In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do **not** fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.
  1. **Qualitative/educational spectra.** The simulated spectra serve to recognise functional groups and understand structure-spectrum relationships, not to replace certified experimental data.
  1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.
  1. **Two-way accuracy.** For rigorous use, direct links to the official databases (NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.
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## 6. Release acceptance criteria

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-  `node --check` passes without errors on the modified modules.
  -  Playwright E2E: 0 non-network JS errors.
  -  No visible regression in the touched sections.
  -  Service Worker cache version bumped.
  -  `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
  -  `verifica_guida`: whatever the documentation promises exists in the code.
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## 7. Verification battery — state at version `bsi-v181`

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Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

### 7.1 Composition

| Family          | Benches                                                                                                                                                                   | Subject                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Scientific data | <code>verifica-farmaci</code> , <code>test_spettri</code> , <code>test_assi</code> , <code>test_assi_canvas</code> , <code>test_costanti</code> , <code>audit_dati</code> | structure vs molecular weight, functional-group recognition, axis conventions, physical constants |

| Family      | Bench                                                                                                                        | Subject                                                                                                                             |
|-------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| AI agent    | test_ko, test_404, test_503, test_firma, test_attesa, test_attesalunga, test_tetto, test_nucleo, browser_ko, browser_prova   | retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers                     |
| Stability   | audit_stabilita, audit_promesse, audit_quota, test_filemanager, test_doppioinvio, test_sw, test_visore3d                     | long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts |
| Interface   | browser_reset, browser_proxy, browser_proxyui, browser_rdkit, browser_lab, browser_frontiera, test_aggiorna, test_guidaproxy | panels, proxy, RDKit laboratory, updates                                                                                            |
| Consistency | verifica_guida                                                                                                               | documentation ↔ code correspondence, conflict markers across 58 files                                                               |

## 7.2 Properties verified that a functional test does not observe

| Property                           | Method                                                    | Measured result                                                                  |
|------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------|
| Absence of memory leaks            | 89 sections opened for 5 rounds, DOM node count per round | construction +26,876 nodes on the first round, <b>+0 on the four that follow</b> |
| Survival of exhausted storage      | localStorage.setItem forced to throw                      | 10 pages out of 10 stay operational                                              |
| Network degradation (≠ no network) | requests held for 20 s, interception at context level     | answer served from cache in <b>3,507 ms</b> (threshold 3,500 ms)                 |
| Spectrum reproducibility           | drawn twice, compared byte for byte                       | identical                                                                        |
| WebGL contexts                     | viewer constructions counted across 10 molecules          | from 7 down to <b>1</b>                                                          |

## 7.3 Notes on method

**A bench that measures nothing passes**, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.
- Probing `localStorage` exhaustion with a one-byte write proves nothing: after saturation, one byte always finds room.
- The ticks of an axis drawn on `<canvas>` are not readable as text: they must be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.